Inertia · Frayer & doodle notes

Name _____ Partner _____

SECTION A BUILD THE WORD

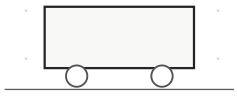
Focus word **Inertia**

From Section A: Inertial frame, Net force, Inertia, Equilibrium.

Meaning in my words What does inertia mean? Say it the way you'd explain it to a friend.	**How can you tell?** An empty cart and a loaded cart both start at rest. We push each with the same net force for one second. Which gains less speed? Why?
Example or sketch A cart is already rolling right. The forces on it balance. Where would you draw it one second later? And one second after that?	**Non-example and why** Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them?

SECTION B DOODLE THE IDEA

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

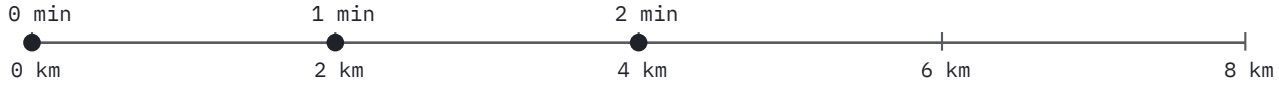


My marks mean _____

Where should we point the telescope?

SECTION E / FINAL MEETING Name _____

The telescope spots XJ once a minute. At minute 3, the camera can take one picture. Your team gets to choose where it points.



Camera windows A +4 km B +6 km C +8 km

Choose a job together. One of you marks the prediction. The other makes the call.

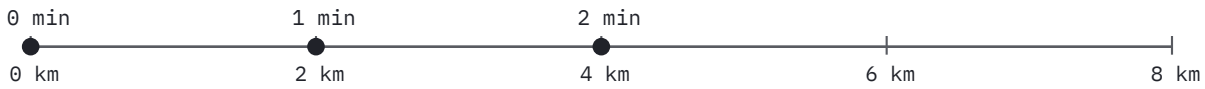
☐ **Tracking engineer**

Circle the window you'd aim for. Tell the camera operator where to point and which two sightings helped you decide.

☐ **Live science broadcaster**

Tell viewers where to look. Someone asks, "If XJ has no engine, why does it keep moving?" Use your cart drawing to answer.

Mark your prediction here.



WHEN THE NEXT SIGHTING COMES IN

Mark the new sighting. How far did XJ move in the last minute? Switch speakers and tell your crew or your viewers what changed.

Why could a straight-line forecast work for a minute even while gravity pulls on XJ?